

Jiayuan Fan

Contact Information	Fudan University College of Intelligent Robotics and Advanced Manufacturing Yangpu District, Shanghai, P.R.China 200433	<i>Fax:</i> +86-21-65644937 <i>E-mail:</i> jyfan@fudan.edu.cn <i>Web:</i> https://fansarah.github.io/
Current	Fudan University , Shanghai, P. R. China Associate Professor, College of Intelligent Robotics and Advanced Manufacturing, Jan. 2024 – Present Associate Professor, Academy for Engineering and Technology, Mar. 2019 – Dec. 2023	
Previous	Agency for Science, Technology and Research , Singapore Scientist II, Institute for Infocomm Research (I2R), A*STAR, Apr. 2018 – Feb. 2019 Scientist I, Institute for Infocomm Research (I2R), A*STAR, Jan. 2014 – Mar. 2018	
Education	Nanyang Technological University , Singapore Ph.D in Information Engineering, Aug. 2008 – Feb. 2015 Harbin Institute of Technology , Harbin, P.R.China M.Eng in Information and Communication Engineering, Sep. 2006 – Jul. 2008 Harbin University of Science and Technology , Harbin, P.R.China B.Eng in Electronic Information Engineering, Sep. 2002 – Jul. 2006	
Research Interests	Multimodal Learning 3D Scene Reconstruction Hyperspectral Image Classification Autonomous Driving Perception Unsupervised/Weakly Supervised Learning	
Honors and Awards	Shanghai Thousand Talents Program (Long-term Innovation) , 2020 Shanghai Pujiang Talent Program, 2019 CCF-Tencent Rhino-Bird Program, Tencent Technology (Shenzhen) Co., Ltd., 2024 First Prize, Teaching Innovation Competition of Fudan University, Fudan University, 2024 Third Place, Anomalous Road Object Detection Challenge, ECCV 2022 Winner Award, Infrared Small Target Detection Challenge under Resource Constraints, ICPR 2024 Third Prize, “Excellence Cup” Innovation and Entrepreneurship Competition, Fudan University, 2024 Third Prize, China Graduate AI Innovation Competition, Association of Chinese Graduate Education, 2023 Third Prize, The China Graduate Electronics Design Contest, China Electronics Association, 2024	

Grants

PI: Variational Inference Based Time-sensitive 3D Reconstruction Theory and Method for Outdoor Dynamic Objects, National Natural Science Foundation of China, P. R. China, 2026-2029, ¥ 500,000

PI: Hierarchical Self-supervised Landmark Recognition Method from the Perspective of Unmanned Aerial Vehicles, National Natural Science Foundation of China, P. R. China, 2022-2024, ¥ 300,000

PI: Land Use Monitoring Technology Based on High-resolution Remote Sensing Satellite Images, Shanghai Municipal Science and Technology Commission, P. R. China, 2019-2021, ¥ 300,000

PI: Industrial and Residential Space Intelligent Technology School-Enterprise Joint Research Center, Industry-University Joint Laboratory, P. R. China, 2024-2027, ¥ 15,000,000

PI: Development of Spatial Intelligence Multimodal Models, Shanghai Moushen Intelligent Technology Co., Ltd., P. R. China, 2026-2029, ¥ 5,500,000

PI: An Unsupervised Pedestrian Re-Identification Method, System and Computer Readable Medium, Hangzhou Xinuo Intelligent Medical Co., Ltd., P. R. China, 2024-2041, ¥ 200,000

PI: Multimodal 3D Scene Embodied Large Model Technology, China Computer Federation, P. R. China, 2024-2025, ¥ 150,000

PI: High-precision, High-efficiency, Comprehensive Store-wide Detection, Tracking and Modeling, Beijing Xiaomi Mobile Software Co., Ltd., P. R. China, 2024-2025, ¥ 700,000

PI: SLAM-based Indoor and Outdoor Spatial Modeling and Recognition with Unity Service Technology Development, Shanghai Luoman Technology Co., Ltd., P. R. China, 2024-2025, ¥ 745,000

Sub-PI: Foundational Machine Learning Algorithms for Generic Computer Vision, Ministry of Science and Technology, P. R. China, 2024-2025, ¥ 400,000

KP: Distributed Autonomous Learning Theory and Methods for Networked Intelligent Manufacturing , National Natural Science Foundation of China (Key Joint Program), P. R. China, 2020-2023, ¥ 210,000

KP: Research on Multi-modal Fusion and Knowledge Reasoning under Limited Computing Power and Data Resources , Shanghai Municipal Natural Science Foundation (General Program), P. R. China, 2023-2026, ¥ 65,000

KP: Statistical Explainable Cross-Media Small Sample Deep Learning Algorithms , Key Laboratory Foundation, P. R. China, 2022-2023, ¥ 170,000

KP: General Visual Dataset and Evaluation Platform , National Key R&D Program of China, P. R. China, 2023-2025, ¥ 300,000

KP: Noise-Robust Optical Matrix Computation for Low-Precision Quantization Adapted to Transformer Models, Shanghai Science and Technology Commission (Explorer Program), P. R. China, 2024-2025, ¥ 500,000

KP: Frontier Basic Theories and Key Technologies of Artificial Intelligence , Shanghai Municipal Major Science and Technology Project, P. R. China, 2020-2023, ¥ 300,000

Selected Journal Publications

1. Chenchen Wang, Jiamu Sheng, Jingyi Zhou, Zhende Song, **Jiayuan Fan** and Zhouchen Lin, AdaMixCon: Difficulty-Aware Feature Enhancement for Few-Shot Hyperspectral Image Classification, IEEE Transactions on Geoscience and Remote Sensing (*IEEE TGRS*), 64:1-15, 2026
2. Jingdong Zhang, **Jiayuan Fan**, Peng Ye, Bo Zhang, Hancheng Ye, Baopu Li, Yancheng Cai, Tao Chen, BridgeNet: Comprehensive and Effective Feature Interactions via Bridge Feature for Multi-Task Dense Predictions, IEEE Transactions on Pattern Analysis and Machine Intelligence (*IEEE TPAMI*), 47(5):3657-3672, 2025.
3. Jiamu Sheng, Jingyi Zhou, Jiong Wang, Peng Ye, Jiayuan Fan, DualMamba: A Lightweight Spectral-Spatial Mamba-Convolution Network for Hyperspectral Image Classification, IEEE Transactions on Geoscience and Remote Sensing (*IEEE TGRS*, *ESI Highly Cited*), 63:1-15, 2025.
4. Linyi Zhou, Zhongxue Gan, **Jiayuan Fan**, Centercoop: Center-based Feature Aggregation for Communication-efficient Vehicle-infrastructure Cooperative 3D Object Detection, IEEE Robotics and Automation Letters (*IEEE RAL*), 9(4):3570-3577, 2024.
5. Zhen Mei, Peng Ye, Baopu Li, Tao Chen, **Jiayuan Fan**, Wanli Ouyang, DeNKD: Decoupled Non-Target Knowledge Distillation for Complementing Transformer-Based Unsupervised Domain Adaptation, IEEE Transactions on Circuits and Systems for Video Technology (*IEEE TCSVT*), 34(5):3220-3231, 2024.
6. Lin Zhan, Peng Ye, **Jiayuan Fan**, Tao Chen, U2ConvFormer: Marrying and Evolving Nested U-Net and Scale-Aware Transformer for Hyperspectral Image Classification, IEEE Transactions on Geoscience and Remote Sensing (*IEEE TGRS*), 62:1-14, 2024.
7. Jingyi Zhou, Jiamu Sheng, Peng Ye, **Jiayuan Fan**, Tong He, Bin Wang, Tao Chen, Exploring Multi-Timestep Multi-Stage Diffusion Features for Hyperspectral Image Classification, IEEE Transactions on Geoscience and Remote Sensing (*IEEE TGRS*, *ESI Highly Cited*), 62:1-16, 2024.
8. Lin Zhang, Bo Zhang, Botian Shi, **Jiayuan Fan**, Tao Chen, Few-Shot Cross-Domain Object Detection With Instance-Level Prototype-Based Meta-Learning, IEEE Transactions on Circuits and Systems for Video Technology (*IEEE TCSVT*), 34(10):9078-9089, 2024.
9. Mingsheng Li, Lin Zhang, Mingzhen Zhu, Zilong Huang, Gang Yu, **Jiayuan Fan**, Tao Chen, Lightweight Model Pre-Training via Language Guided Knowledge Distillation, IEEE Transactions on Multimedia (*IEEE TMM*), 26:10720-10730, 2024.
10. Wanzhang Li, Fukun Yin, Wen Liu, Yiyang Yang, Xin Chen, Biao Jiang, Gang Yu, **Jiayuan Fan**, Unbounded-GS: Extending 3D Gaussian Splatting With Hybrid Representation for Unbounded Large-Scale Scene Reconstruction, IEEE Robotics and Automation Letters (*IEEE RAL*), 9(12):11529-11536, 2024.
11. Jiaxiang Chen, **Jiayuan Fan**, Hancheng Ye, Jie Li, Yongbing Liao, Tao Chen, Exploring Kernel-Based Texture Transfer for Pose-Guided Person Image Generation, IEEE Transactions on Multimedia (*IEEE TMM*), 25:7337-7349, 2023.
12. Ke Wu, **Jiayuan Fan**, Peng Ye, Mingzhen Zhu, Hyperspectral Image Classification Using Spectral-Spatial Token Enhanced Transformer With Hash-Based Positional Embedding, IEEE Transactions on Geoscience and Remote Sensing (*IEEE TGRS*), 61:1-16, 2023.
13. Hancheng Ye, Bo Zhang, Tao Chen, **Jiayuan Fan**, Bin Wang, Performance-Aware Approximation of Global Channel Pruning for Multitask CNNs, IEEE Transactions on Pattern Analysis and Machine Intelligence (*IEEE TPAMI*), 45(8): 10267-10284,

2023.

14. Mingzhen Zhu, **Jiayuan Fan**, Qihang Yang, Tao Chen, SC-EADNet: A Self-Supervised Contrastive Efficient Asymmetric Dilated Network for Hyperspectral Image Classification, *IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS)*, 60:1-17, 2022.
15. Bo Zhang, Hancheng Ye, Gang Yu, Bin Wang, Yike Wu, **Jiayuan Fan**, Tao Chen, Sample-Centric Feature Generation for Semi-Supervised Few-Shot Learning, *IEEE Transactions on Image Processing (IEEE TIP)*, 31:2309-2320, 2022.
16. Yongbin Liao, Hongyuan Zhu, Yanggang Zhang, Chuanguan Ye, Tao Chen, **Jiayuan Fan**, Point Cloud Instance Segmentation With Semi-Supervised Bounding-Box Mining, *IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI)*, 44(12):10159-10170, 2022.
17. Peng Ye, Baopu Li, Tao Chen, Jiayuan Fan, Zhen Mei, Chen Lin, Chongyan Zuo, Qinghua Chi, Wanli Ouyang, Efficient Joint-Dimensional Search with Solution Space Regularization for Real-Time Semantic Segmentation, *International Journal of Computer Vision (IJCV)*, 130(11):2674-2694, 2022.
18. Bo Zhang, Tao Chen, Bin Wang, Xiaofeng Wu, Liming Zhang, Jiayuan Fan, Densely Semantic Enhancement for Domain Adaptive Region-Free Detectors, *IEEE Transactions on Circuits and Systems for Video Technology (IEEE TCSVT)*, 32(3):1339-1352, 2022.
19. Tao Chen, Shijian Lu, **Jiayuan Fan**, SS-HCNN: Semi-Supervised Hierarchical Convolutional Neural Network for Image Classification(*IEEE TIP*), 28(5): 2389-2398, 2019.
20. Tao Chen, Shijian Lu, **Jiayuan Fan**, S-CNN: Subcategory-Aware Convolutional Networks for Object Detection, *IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI)*, 40(10): 2522-2528, 2018.
21. **Jiayuan Fan**, Tao Chen, Shijian Lu, Superpixel Guided Deep-Sparse-Representation Learning for Hyperspectral Image Classification, *IEEE Transactions on Circuits and Systems for Video Technology (IEEE TCSVT)*, 28(11): 3163-3173, 2018.
22. **Jiayuan Fan**, Tao Chen, Shijian Lu, Unsupervised Feature Learning for Land-Use Scene Recognition, *IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS)*, 55(4): 2250-2261, 2017.
23. **Jiayuan Fan**, Hong Cao, Alex C. Kot, Estimating EXIF Parameters Based on Noise Features for Image Manipulation Detection. *IEEE Transactions Information on Forensics and Security*, 8(4): 608-618, 2013.

Selected Conference Publications

1. Zhende Song, Chenchen Wang, Jiamu Sheng, Chi Zhang, Shengji Tang, **Jiayuan Fan**, Tao Chen, DreamFrame: Enhancing Video Understanding via Automatically Generated QA and Style-Consistent Keyframes, *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM)*, 2025.
2. Yiying Yang, Fukun Yin, **Jiayuan Fan**, Xin Chen, Wanzhang Li, Gang Yu, Scene123: One Prompt to 3D Scene Generation via Video-Assisted and Consistency-Enhanced MAE, *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM)*, 2025.
3. Biao Jiang, Xin Chen, Chi Zhang, Fukun Yin, Zhuoyuan Li, Gang Yu, **Jiayuan Fan**, Motionchain: Conversational Motion Controllers via Multimodal Prompts, *Proceedings of the 18th European Conference on Computer Vision (ECCV)*, pp.54-74, 2025.
4. Yiying Yang, Fukun Yin, Wen Liu, **Jiayuan Fan**, Xin Chen, Gang Yu, Tao Chen,

PM-INR: Prior-Rich Multi-Modal Implicit Large-Scale Scene Neural Representation, Proceedings of the 38th AAAI Conference on Artificial Intelligence (*AAAI, Oral*), 38(7):6594-6602, 2024.

5. Sijin Chen, Xin Chen, Chi Zhang, Mingsheng Li, Gang Yu, Hao Fei, Hongyuan Zhu, **Jiayuan Fan**, Tao Chen, LL3DA: Visual Interactive Instruction Tuning for Omni-3D Understanding, Reasoning, and Planning, Proceedings of 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (*CVPR*), pp.26428-26438, 2024.
6. Chong Yu, Tao Chen, Zhongxue Gan, **Jiayuan Fan**, Spear: Evaluate the Adversarial Robustness of Compressed Neural Models, Proceedings of the 33rd International Joint Conference on Artificial Intelligence (*IJCAI*), pp.1598-1606, 2024.
7. Shijie Chen, Mohammad Mahdizadeh, Chong Yu, **Jiayuan Fan**, Tao Chen, Through the Real World Haze Scenes: Navigating the Synthetic-to-Real Gap in Challenging Image Dehazing, Proceedings of 2024 International Conference on Robotics and Automation (*ICRA*), pp.7265-7272, 2024.
8. Chongshan Lu, Fukun Yin, Xin Chen, Wen Liu, Tao Chen, Gang Yu, **Jiayuan Fan**, A Large-Scale Outdoor Multi-Modal Dataset and Benchmark for Novel View Synthesis and Implicit Scene Reconstruction, Proceedings of 2023 IEEE/CVF International Conference on Computer Vision (*ICCV*), pp.7557-7567, 2023.
9. Chong Yu, Tao Chen, Zhongxue Gan, **Jiayuan Fan**, Boost Vision Transformer With GPU-Friendly Sparsity and Quantization, Proceedings of 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (*CVPR*), pp.22658-22668, 2023.
10. Ben Sha, Baopu Li, Tao Chen, **Jiayuan Fan**, Tao Sheng, Rethinking Pseudo-Label-Based Unsupervised Person Re-ID with Hierarchical Prototype-based Graph, Proceedings of the 31st ACM International Conference on Multimedia (*ACM MM*), pp.2090-2100, 2023.
11. Yuhan Ding, Fukun Yin, **Jiayuan Fan**, Hui Li, Xin Chen, Wen Liu, Chongshan Lu, Gang Yu, Tao Chen, PDF: Point Diffusion Implicit Function for Large-scale Scene Neural Representation, Proceedings of the 37th Conference on Neural Information Processing Systems (*NeurIPS*), pp.84-96, 2023.
12. Chuangguan Ye, Hongyuan Zhu, Yongbin Liao, Yanggang Zhang, Tao Chen, **Jiayuan Fan**, What Makes for Effective Few-Shot Point Cloud Classification?, Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision (*WACV*), pp.1829-1838, 2022.
13. Bo Zhang, Jiakang Yuan, Baopu Li, Tao Chen, **Jiayuan Fan**, Botian Shi, Learning Cross-Image Object Semantic Relation in Transformer for Few-Shot Fine-Grained Image Classification, Proceedings of the 30th ACM International Conference on Multimedia (*ACM MM*), pp.2135-2144, 2022.
14. Yongbin Liao, Hongyuan Zhu, Tao Chen, **Jiayuan Fan**, Sperc: Semi-supervised Point Cloud Instance Segmentation With Perturbation Consistency Regularization, Proceedings of 2021 IEEE International Conference on Image Processing (*ICIP*), pp.3113-3117, 2021.
15. Wanling Fan, **Jiayuan Fan**, Gang Yu, Bin Fu, Tao Chen, HSEGAN: Hair Synthesis and Editing Using Structure-Adaptive Normalization on Generative Adversarial Network, Proceedings of 2021 IEEE International Conference on Image Processing (*ICIP*), pp.1324-1328, 2021.
16. Wei Li, **Jiayuan Fan**, Yanwei Fu, Global-to-Local Dynamic Feature Aggregation for Unsupervised Person Re-Identification, Proceedings of 2021 IEEE International Conference on Multimedia and Expo (*ICME*), pp.1-6, 2021.
17. Yike Wu, Bo Zhang, Gang Yu, Weixi Zhang, Bin Wang, Tao Chen, **Jiayuan Fan**, Object-aware Long-short-range Spatial Alignment for Few-Shot Fine-Grained Image Classification, Proceedings of the 29th ACM International Conference on Multimedia (*ACM MM*), pp.107-115, 2021.

18. Qihang Yang, Tao Chen, **Jiayuan Fan**, Ye Lu, Chongyan Zuo, Qinghua Chi, EADNet: Efficient Asymmetric Dilated Network for Semantic Segmentation, Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing (*ICASSP*), pp.2315-2319, 2021.
19. Jingjing Chen, Jichao Zhang, Enver Sangineto, Tao Chen, **Jiayuan Fan**, Nicu Sebe, Coarse-to-Fine Gaze Redirection with Numerical and Pictorial Guidance, Proceedings of 2021 IEEE/CVF Winter Conference on Applications of Computer Vision (*WACV*), pp.3665-3674, 2021.

Patents

1. **Jiayuan Fan**, Wei Li, and Yanwei Fu, “Unsupervised Person Re-identification Method Based on Global-Local System,” CN patent filed 03/2021, issued 05/2023.
2. Chuanguan Ye, Tao Chen, Yanggang Zhang, Yongbin Liao, and **Jiayuan Fan**, “Real-World Domain 3D Point Cloud Object Recognition Method Based on Hierarchical Attention Sampling Strategy,” CN patent filed 08/2021, issued 04/2024.
3. Tao Chen, Weixi Zhang, and **Jiayuan Fan**, “Multimodal Cross-View Image Generation Method Based on Residual-Based Cascaded Progressive Optimization,” CN patent filed 10/2021, issued 07/2025.
4. **Jiayuan Fan**, Tao Chen, and Ke Wu, “Cross-View Image Translation Method Based on Generative Adversarial Networks,” CN patent filed 11/2021, issued 05/2025.
5. **Jiayuan Fan**, Hui Li, and Tao Chen, “Passive Domain Noise-Aware Domain Adaptive Segmentation Method for Remote Sensing Images,” CN patent filed 12/2023.
6. **Jiayuan Fan**, Haixiang Chen, and Tao Chen, “A Depth Estimation Method Based on Uncertainty Multi-Sensor Fusion,” CN patent filed 12/2023.
7. **Jiayuan Fan**, Lin Zhan, and Yaoxin Yang, “Infrared Small Target Detection Method Based on Label Evolution and Global-Local System,” CN patent filed 11/2024.
8. “Diffusion-Based High-Precision Hyperspectral Image Classification Software with Low Annotation Requirement V1.0,” CN software copyright registered 11/2024.

Membership

Senior Member, IEEE, 2025
Member, Association for Computing Machinery (ACM)
Member, Young Women Scientists Club of Chinese Institute of Electronics
Member, China Society of Image and Graphics
Member, China Computer Federation (CCF)
Member, IEEE Signal Processing Society
Member, IEEE Geoscience and Remote Sensing Society

Service for Refereed Journals

Reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence
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Reviewer for IEEE Transactions Information on Forensics and Security
Reviewer for IEEE Robotics and Automation Letters
Reviewer for Pattern Recognition

Service for Refereed Conferences

Reviewer for top international conferences including CVPR, ICCV, ECCV, ACM MM, AAAI

Teaching
Experience

Fudan University, P.R.China

Image Processing, 2020–2026

Machine Learning, 2019–2026

Computational Neuroscience, 2022-2026